

Regional Economic Integration

區域經濟整合

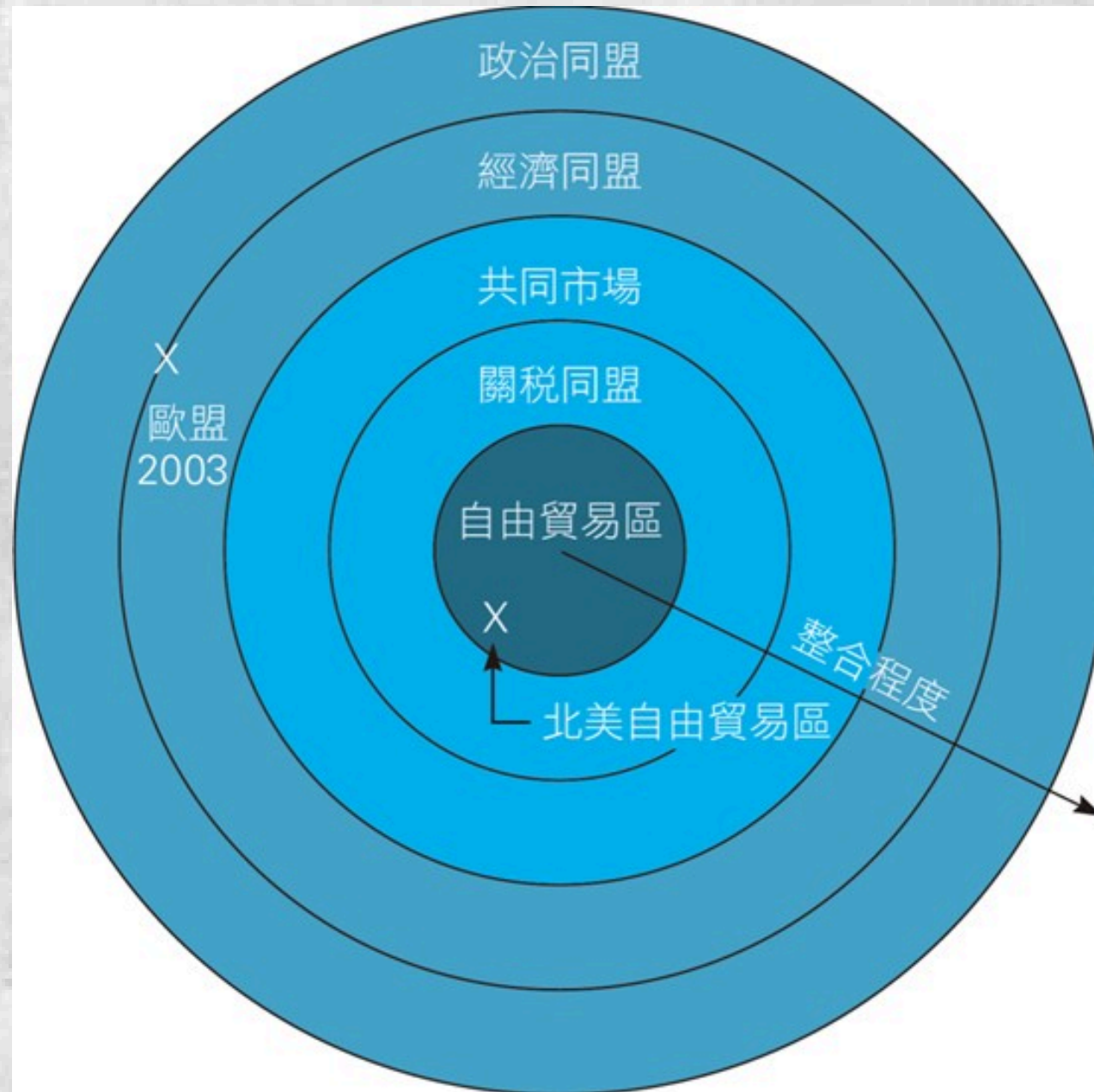
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Regional Economic Integration refers to **agreements** between countries in a **geographic region** to reduce or eliminate trade barriers among themselves, while possibly maintaining barriers against non-members. The main objective is to increase **economic efficiency**, **boost trade**, **attract investment**, and **enhance bargaining power** in global negotiations.

Levels of Regional Economic Integration

Level	Description	Tariff on Members	Common External Tariff	Free Movement of Factors	Examples
Preferential Trade Agreement (PTA)	Reduced tariffs on selected goods	Reduced	No	No	Many developing country deals
Free Trade Area (FTA)	Elimination of tariffs on substantially all trade	Zero	No (each keeps own)	No	USMCA, ASEAN FTA, RCEP
Customs Union (CU)	FTA + Common External Tariff	Zero	Yes	No	Mercosur, Southern African CU
Common Market	Customs Union + Free movement of labor & capital	Zero	Yes	Yes	EU Single Market (earlier stage)
Economic & Monetary Union	Common Market + Common currency + coordinated policies	Zero	Yes	Yes	Eurozone (EU)
Political Union	Full integration with Common Government	Zero	Yes	Yes	(Theoretical)



Static vs Dynamic Effects

Type	Effects	Usually More Important?
Static Effects	Trade creation, diversion, terms of trade	Short to medium term
Dynamic Effects	Economies of scale, competition, technology transfer, investment, innovation	Long term (often larger)

Dynamic Benefits:

- Larger market → economies of scale, Increased competition → efficiency gains (X-efficiency)
- Attracts Foreign Direct Investment (FDI), Better infrastructure and policy coordination.

Pros (Advantages) of Regional Integration

Category	Advantages	Explanation
Economic	Trade Creation	Replaces high-cost domestic production with cheaper partner production
	Economies of Scale	Larger regional market allows firms to produce more efficiently
	Increased Competition	Forces domestic firms to become more efficient ("X-efficiency")
	Investment Attraction (FDI)	Larger market + policy stability attracts foreign investors
	Technology Transfer & Innovation	Easier movement of people, capital, and knowledge
	Better Infrastructure & Policy Coordination	Joint projects in transport, energy, standards
Political	Enhanced Bargaining Power	Larger bloc has more influence in WTO or with big powers (e.g., EU, ASEAN)
	Peace and Stability ("Peace Dividend")	Countries that trade intensively are less likely to fight (EU example)
	Domestic Reform Anchor	International commitments help lock in difficult domestic reforms
Strategic	Supply Chain Resilience	Regional integration reduces dependence on distant suppliers
	Geopolitical Influence	Creates counterweight to global powers

Cons (Disadvantages)

Category	Disadvantages	Explanation
Economic	Trade Diversion	May shift imports from efficient global producers to less efficient partners
	Loss of Tariff Revenue	Important revenue source for many developing countries
	Adjustment Costs & Unemployment	Some industries contract, leading to job losses in the short term
	Risk of "Race to the Bottom"	Pressure to lower taxes, labor, or environmental standards
	Uneven Distribution of Gains	Some regions/industries benefit much more than others (e.g., core vs periphery)
Sovereignty	Loss of Policy Autonomy	Must follow common rules (tariffs, regulations, subsidies)
	Harmonization Costs	Expensive to align standards, laws, and regulations
Political	Increased Complexity & Bureaucracy	New institutions and decision-making processes
	Risk of Domination by Larger Members	Smaller countries may lose influence (e.g., concerns in Mercosur, AfCFTA)
Other	Exclusion of Non-Members ("Spaghetti Bowl" effect)	Proliferation of overlapping FTAs creates complex rules of origin
	Potential for Conflict	Disputes over rules, compensation, or external tariffs

Major Regional Trade Agreements (2026)

- **EU** – Deepest integration (Economic & Monetary Union)
- **USMCA** – Modern FTA (USA, Mexico, Canada)
- **RCEP** – Largest by population (ASEAN + China, Japan, Korea, Australia, NZ)
- **CPTPP** – High-standard FTA (11 countries, including Japan, Vietnam, Canada)
- **AfCFTA** – African Continental Free Trade Area (ambitious but implementation ongoing)
- **ASEAN Economic Community**

Key Economic Concepts

Trade Creation vs Trade Diversion (Jacob Viner, 1950)

- **Trade Creation:** Replacement of `high-cost domestic production` with `lower-cost imports` from partner countries → **Welfare Improving.**
- **Trade Diversion:** Replacement of `low-cost imports` from the rest of the world with `higher-cost imports` from partner countries (due to preferences) → **Welfare Reducing.**

Model Setup (Small Country)

Assume a small country forms an **FTA** with a partner country.

- **Demand:** $Q_d = D(P)$, **Supply:** $Q_s = S(P)$
- P_c = World price (most efficient producer), P_b = Partner country price ($P_c < P_b$)
- t = Specific tariff rate

Prices:

- Pre-FTA import price (from world) = $P_c + t$
- Post-FTA import price from partner = P_b (tariff removed)
- Post-FTA import price from world = $P_c + t$ (tariff remains)

Key Equations

Before FTA (with tariff on all imports):

$$P_1 = P_c + t, \quad \text{Imports} = D(P_1) - S(P_1)$$

After FTA:

$$P_2 = P_b \quad (\text{assuming } P_c + t > P_b > P_c), \quad \text{Imports from Partner} = D(P_2) - S(P_2)$$

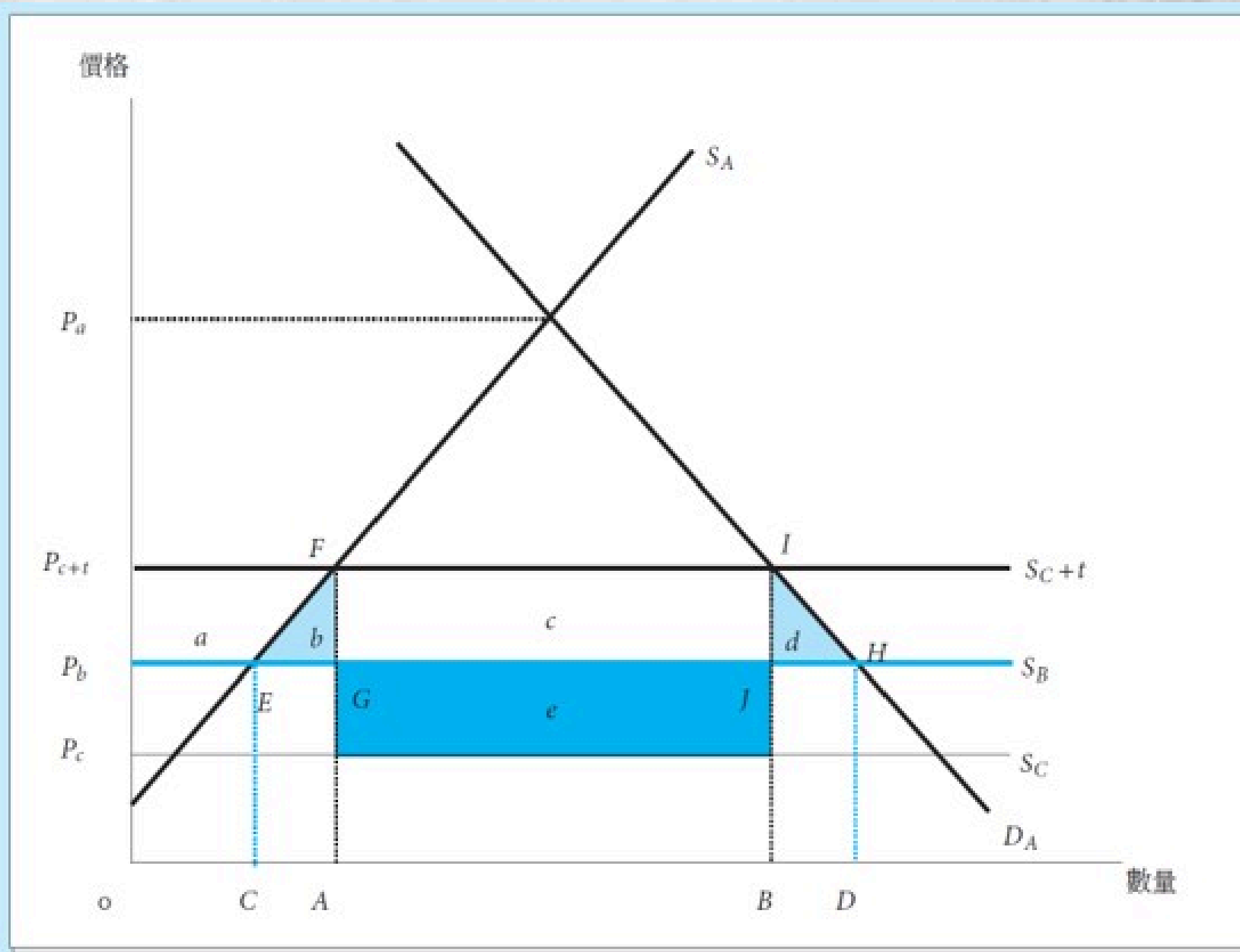
- **Trade Creation** = Increase in total imports due to lower price --> mainly replaces domestic poor production
- **Trade Diversion** = Portion of imports that switches from world to partner

Welfare Diagram Description

Imagine the diagram with Price on vertical axis, Quantity on horizontal axis.

Lines:

- Domestic Demand (downward sloping), Domestic Supply (upward sloping)
- Horizontal lines for prices: P_c , P_b , $P_c + t$



Welfare Areas (Detailed Labeling)

When the country moves from $P_c + t$ to P_b :

- **Consumer Surplus Gain** = Area (a + b + c + d),
- **Producer Surplus Loss** = Area a
- **Government Tariff Revenue Loss** = Area (c + e)
 - lost on imports now coming from partner

- **Net Welfare Effect** = $(b+d)-e$, Where:
 - **Area b = Trade Creation Gain** (production efficiency)
 - **Area d = Trade Creation Gain** (consumption efficiency)
 - **Area e = Trade Diversion Loss**

Interpretation:

- If $(b + d) > e \rightarrow$ Net welfare **gain** from FTA;
- If $(b + d) < e \rightarrow$ Net welfare **loss** from FTA

Summary Table

Effect	Area	Welfare Impact	Meaning
Consumer Gain	$a+b+c+d$	Positive	Lower price
Producer Loss	$-a$	Negative	Reduced output
Lost Tariff Revenue	$-(c+e)$	Negative	Government loss
Trade Creation	$b+d$	Positive	Efficiency gain
Trade Diversion	$-e$	Negative	Inefficient source
Net Welfare	$b+d-e$	Ambiguous	Depends on magnitudes

Dynamic Gains from Regional Economic Integration

- While **static effects** (trade creation vs diversion) are one-time reallocations of resources,
- **Dynamic gains** are the long-term, ongoing benefits that drive **sustained economic growth**. These are generally considered much larger and more important than static effects.

Main Sources of Dynamic Gains

Dynamic Gain	Mechanism	Expected Impact
Economies of Scale	Larger regional market allows firms to increase production scale	Lower average costs, higher competitiveness
Increased Competition	More foreign competitors force firms to reduce inefficiency ("X-inefficiency")	Higher productivity, innovation
Technology Transfer & Spillovers	FDI, imports of capital goods, movement of skilled workers	Faster technological progress
Foreign Direct Investment (FDI)	Bigger market + policy certainty attracts more investment	Capital inflow, job creation, knowledge transfer
Product Variety & Innovation	Larger market supports more product differentiation	Consumer welfare + learning-by-doing
Structural Transformation	Shift of resources from low-productivity to high-productivity sectors	Long-term growth acceleration
Policy Reform & Credibility	External commitments lock in better policies	Better institutions and governance

Economic Theory Behind Dynamic Gains

A. Economies of Scale (Internal & External)

- Firms face decreasing average costs as output increases.
- **Simple Equation:**

$$AC = \frac{TC}{Q} \quad \text{where } AC \text{ falls as } Q \text{ (output) rises in larger market}$$

- **Internal economies:** Firm-level (better machinery, specialization)
- **External economies:** Industry-level (clustering, shared suppliers, skilled labor pools)

B. Pro-Competitive Effect & X-Efficiency (Leibenstein)

Before integration: Protected firms operate with slack (inefficiency). After integration: Competition forces firms closer to the efficiency frontier.

C. Endogenous Growth Theory (Romer, Grossman-Helpman)

Integration increases:

- Knowledge spillovers; Returns to R&D; Rate of innovation

Growth Equation (Simplified):

$$g = f(\text{openness, market size, human capital, R\&D})$$

Larger integrated markets raise the growth rate g .

Numerical / Illustrative Example

Scenario: Country A has a domestic market of 10 million consumers. It forms an FTA with Country B (another 15 million).

- Before Integration:
 - Firm produces 50,000 cars/year → Average Cost = \$22,000
 - Limited competition → operates at 80% efficiency
- After Integration
(25 million combined market):
 - Firm can sell 120,000 cars/year → Average Cost drops to **\$16,000** (economies of scale)
 - Competition rises → efficiency rises to 95%
 - Productivity growth: +2.5% per year (instead of 0.8%)

Cumulative Effect over 10 years: The dynamic gains can be **3-5 times larger** than one-time static gains.

Conditions for Maximizing Dynamic Gains

Dynamic gains are **not automatic**. They depend on:

1. **Complementary Policies:** Education, infrastructure, flexible labor markets, good governance.
2. **Depth of Integration:** Deeper agreements (Common Market > FTA) generate larger dynamic effects.
3. **Initial Conditions:** Countries with similar technological levels gain more from scale and competition.
4. **Openness to the World:** Integration should be “open regionalism” rather than closed blocs.
5. **Adjustment Support:** Retraining programs, social safety nets to manage short-term costs.

Key Takeaway:

- Most economists believe that **dynamic gains** from successful regional integration are the primary reason countries pursue it.
- While static effects can be small or even negative, the long-term benefits from **productivity growth**, **innovation**, and **structural change** often make integration worthwhile – **provided** countries implement supporting domestic policies.